

Further, features like text detection, line descriptors, general use optimisation (hill climbing and linear programming), computational photography algorithms (HDR, inpainting, edge-aware filters and superpixels), tracking and optical flow algorithms, 2D shape-matching module and 3D surface-matching module, RGB-D module and visualisation tool-kit based 3D visualisation module have been introduced.

Installation and setup

The installation process of OpenCV involves installing libraries and packages for different platforms. Although it seems complicated, tutorials on the website guide you with step-by-step details of doing the same. The library can be downloaded from an already existing setup or one can even create his or her own library from scratch.

The latest version OpenCV 3.0 can be downloaded from SourceForge, for various platforms, as follows:

Windows self-extracting archive: sourceforge.net/projects/opencvlibrary/files/opencv-win/3.0.0

iOS framework: sourceforge.net/projects/opencvlibrary/files/opencv-ios/3.0.0

The source code for all platforms can be downloaded from GitHub: github.com/Itseez/opencv/archive/3.0.0.zip

Various books have been authored exclusively on OpenCV. Getting a hand on one might prove to be just what we need to work our way through this software. There are books like *OpenCV for Secret Agents*, *Computer Vision with OpenCV* and *Learning OpenCV*, to name a few. Books even come in platform-specific versions. One can take a look at the books on opencv.org/books.html

What the users have to say

OpenCV seems to be top-notch when one takes into account first-hand experience. Starting from installation to running the tool, the process turns

More about OpenCV

Image to matrix. All images, be it from a camera or scan or even magnetic resonance imagery, are converted to a matrix containing intensity values of pixels, with the help of Mat. OpenCV processes and manipulates the information to perform necessary tasks.

Storing an image. Images can be stored in colour formats as required by the user. The user decides the colour space and data type to be used, for example, grey-scale or red-blue-green (RBG) formats.

The different methods of scanning images, decided by the style of coding are:

1. Efficient way using classic C style
2. Iteration method that considers images as sections and reads these by approximating each section
3. On-the-fly method used when one wants to specify the row and column number of the item that is to be accessed
4. Core function method that creates look-up tables (LUTs) that can be referenced in the code. Data from LUTs is automatically taken into consideration when called upon, and it either replaces or aids calculation of the particulars of the image being processed.

This is what the makers conclude, "If possible, use the already-made functions of OpenCV. The fastest method turns out to be LUT function. However, if you need to write a simple image scan, use pointer method. Iterator is a safer bet, however, it is quite slower. Using on-the-fly reference access method for full image scan is the most costly in debug mode. In the release mode it may beat the iterator approach, but it surely sacrifices the safety trait of iterators for this."

out to be smooth. Easy-to-read documentation, a powerful and well-designed library and great algorithms appeal to the user.

What works in OpenCV's favour is the user-friendly software and multiple features, along with regular updates. Good quality, being stable in terms of functionality and fast working pace make people adapt to it easily.

GPU module is a value addition. Some go on to say that, with just a little improvement on the GUI front and specific documentation sections, the tool could even become the future of image processing. No wonder then that OpenCV has a rating of 4.8 on 5 from SourceForge.

The website provides documentation and tutorials for every possible operation that can be performed within OpenCV. An answers page allows you to post questions or replies to others' questions. Even after repeated attempts, if bugs are



Fig. 3: Analysing a photo using a histogram (Image courtesy: picshype.com)

not solved, you can report these or request new features to be added, on the website. You can find OpenCV projects at code.opencv.org/projects/opencv

If you want to try out other tools, you could go for VLFeat, WEKA or ToolIP.

Try OpenCV to play with your images and make these fit for your application. All you need to do is install the setup from our DVD. Make sure you choose the right version for your platform. Open into the OpenCV experience! ●

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